

Towards Real-Time High Fidelity Dosimetry

A Deep Learning Surrogate for Radiotherapy

Speaker: Asier López Mantecón

Ibon García Gómez

University of Cantabria – Instituto de Física de Cantabria (IFCA)

FlipPhysics – 2026

Presentation Outline



Introduction

The actual clinical situation: Accuracy vs. Speed in radiotherapy planning.



GT Simulation

Monte Carlo framework: OpenGATE code, source modeling, and dataset.



Deep Learning

Neural Network architecture (3D TransUNet + FiLM) and performance metrics.



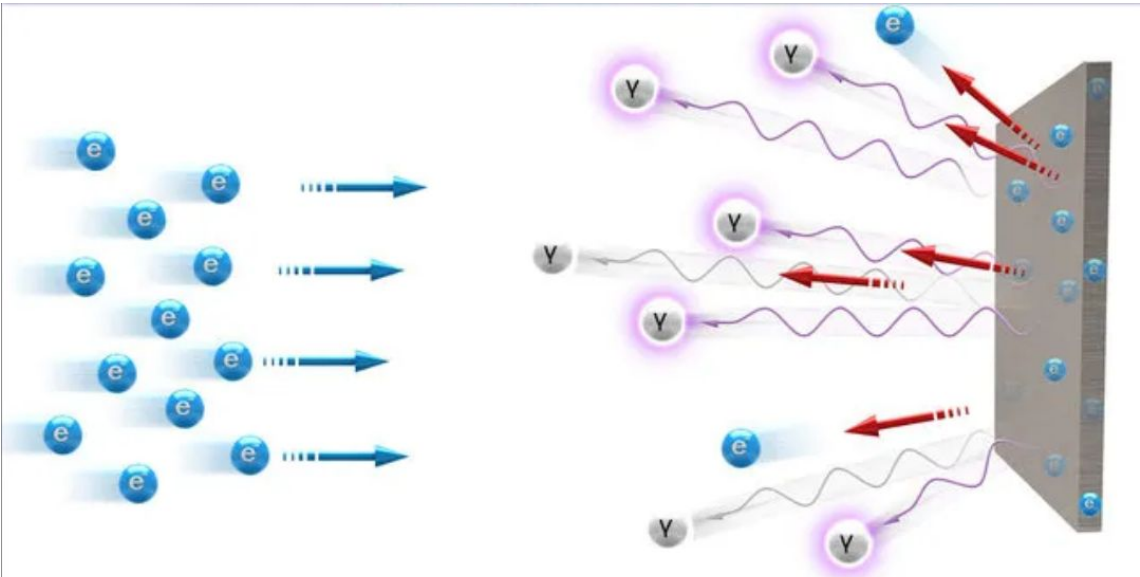
Results

Validation results and path towards differentiable inverse planning.

Introduction: The Current Radiotherapy situation

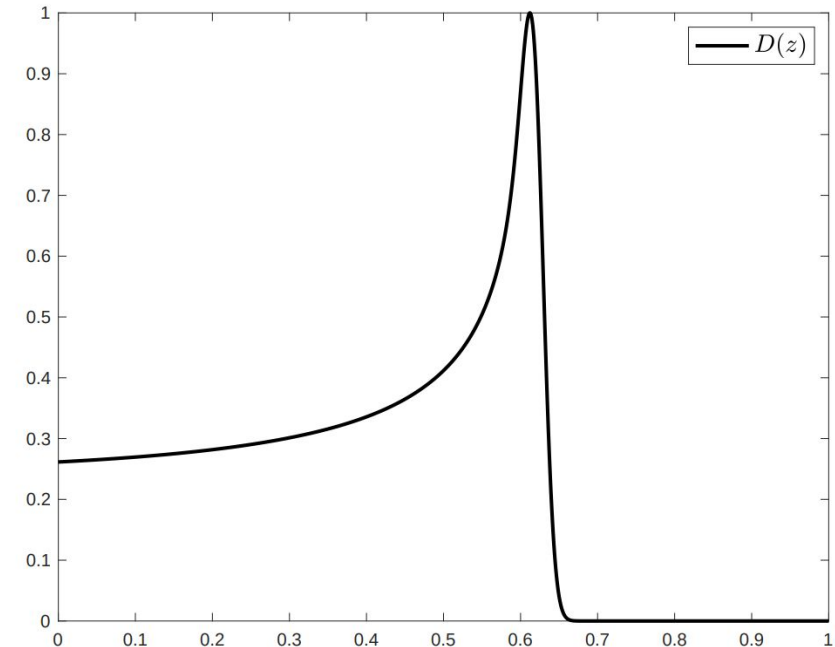
Photon radiated therapies

In most hospitals, there are multiple types of cancer treatments. But the most abundant and versatile one for different types of disease is the radiotherapy with photons.



Protontherapy, The Close Future

The future of treatment plans will be focused on proton or even heavy hadrons. Favourite option for researches and hospitals in the last years.



[Crossley, A. et. al. \(2024\). Jump stochastic differential equations for the characterization of the Bragg peak in proton beam radiotherapy.](#)

Introduction: The Dosimetry Dilemma

The Computational Tax

Monte Carlo (MC) simulations provide the most accuracy in simulations, but require massive computational resources. High-fidelity complex scenarios can take **hours or days** to simulate.

The Clinical Compromise

Current clinical workflows rely on analytical approximations. While fast, they sacrifice precision in tissue interfaces (lung/bone), leading to less accurate delivered doses.

Solution:

A Deep Neural Surrogate that makes the radiotherapy treatments much more fast, enabling real-time treatments.

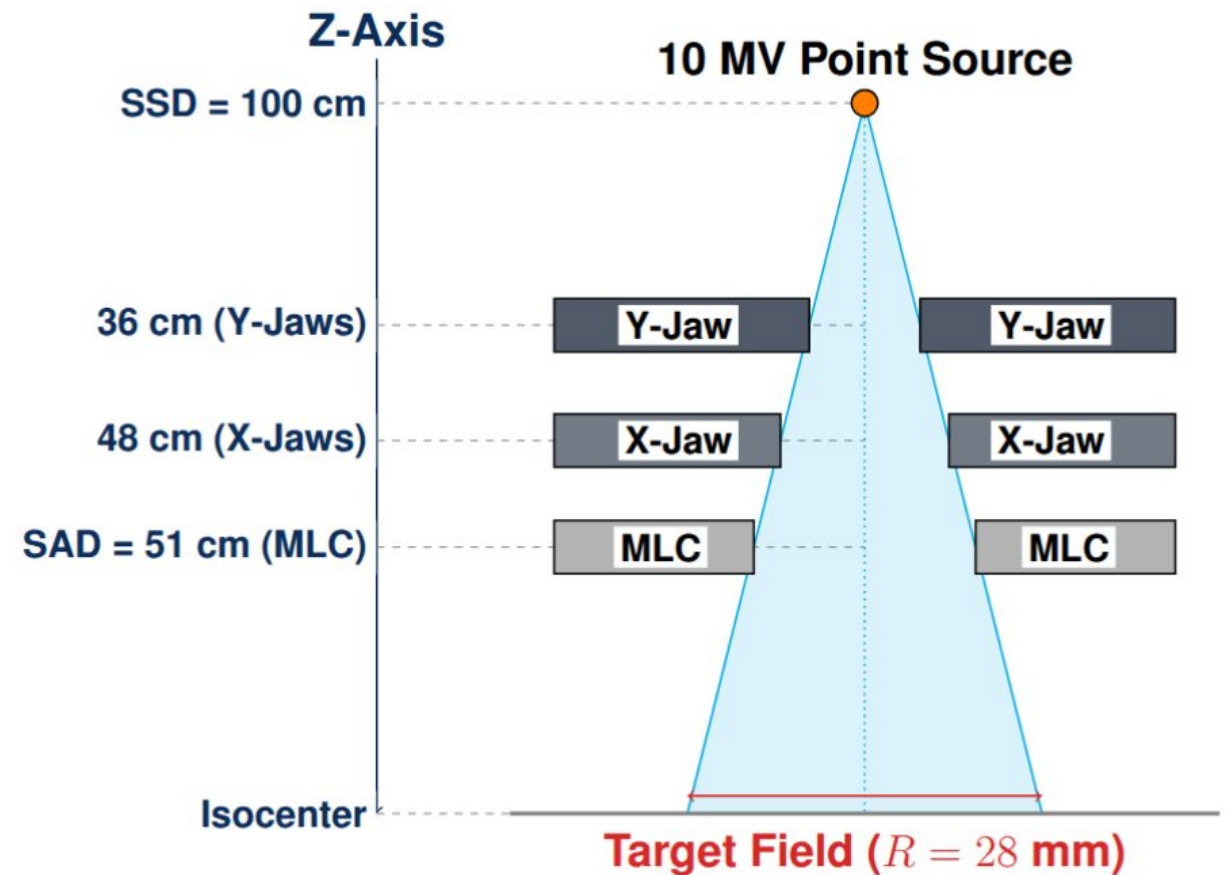
GT Simulations: Source Modeling

Based on real Bremsstrahlung spectrum for photon generation from Mohan et. al.¹

 **Continuous Sampling:** $U(6, 15)$ MV energy range Bremsstrahlung spectrum for photon generation.

 **Circular Source:** Point Source (SSD 100 cm) collimated into a 28 mm radius circular field

 **Mean Energy:** ~ 2.2 MeV for the 10 MV photon fluence.



GT Simulations: Large-Scale Dataset



Patient Phantoms

9 Prostate clinical cases resampled to 3mm isotropic voxels (200 x 200 x 200 grid).



Dataset of GT

~1,600 projections generated for the training of the Neural Network



Primary Statistics

10^6 particles per simulation, for reducing computational time.

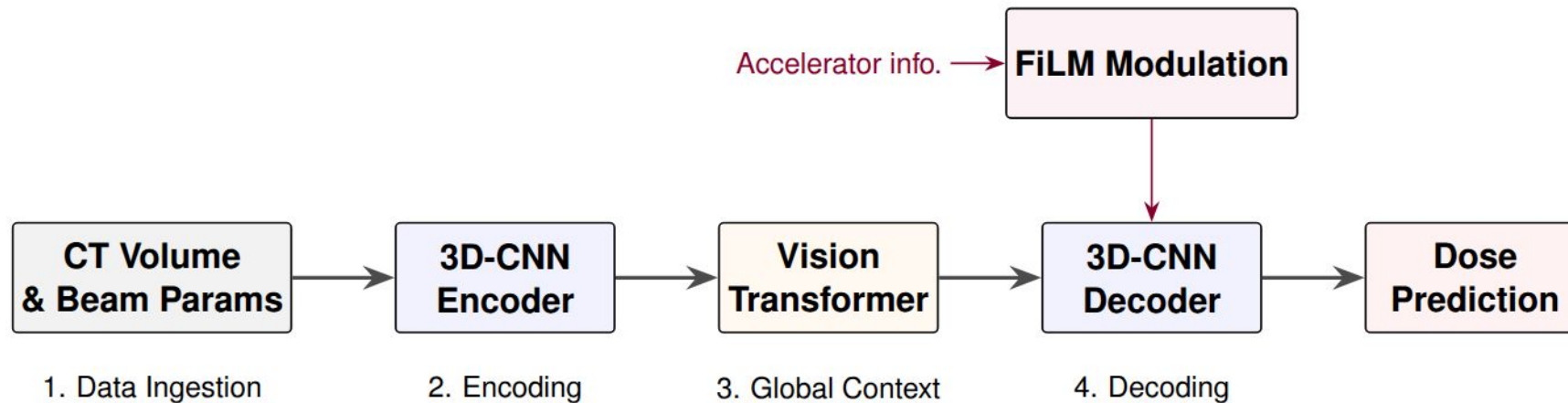
Neural Network Architecture

A hybrid topology designed to balance local anatomical gradients and global dose scatter.

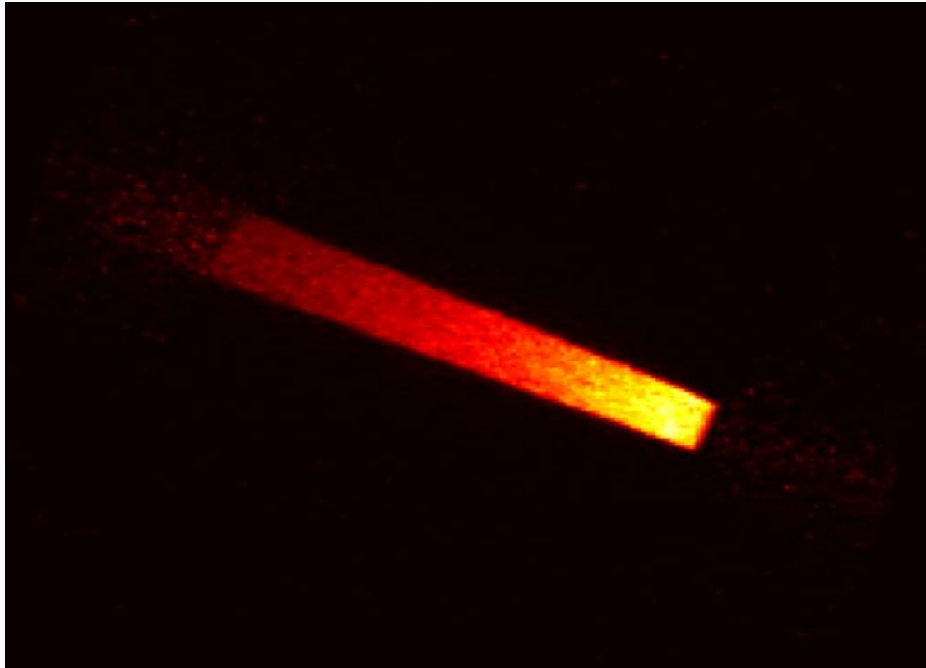
 **3D CNN Encoder:** Learns local density interfaces from CT.

 **Vision Transformer:** Multi-head attention captures long-range dose deposition.

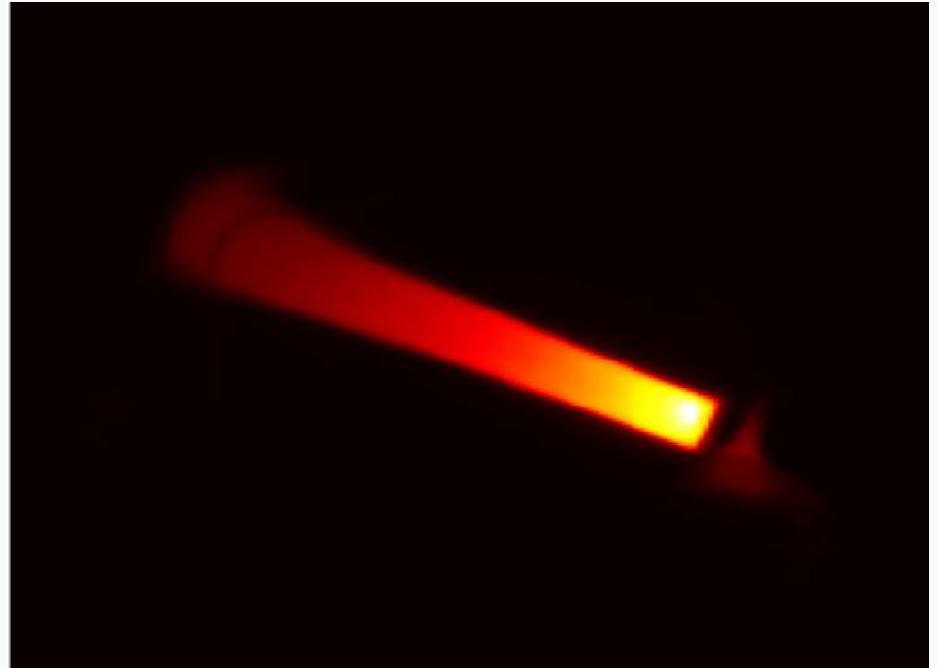
 **FiLM Conditioning:** Injects accelerator properties into the feature maps.



Results: Dose Prediction Fidelity



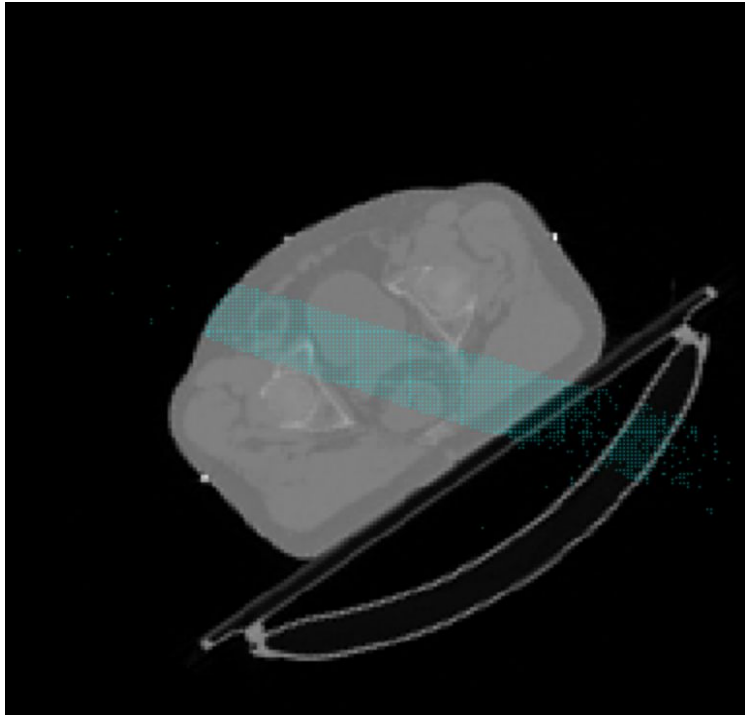
Ground Truth (MC)



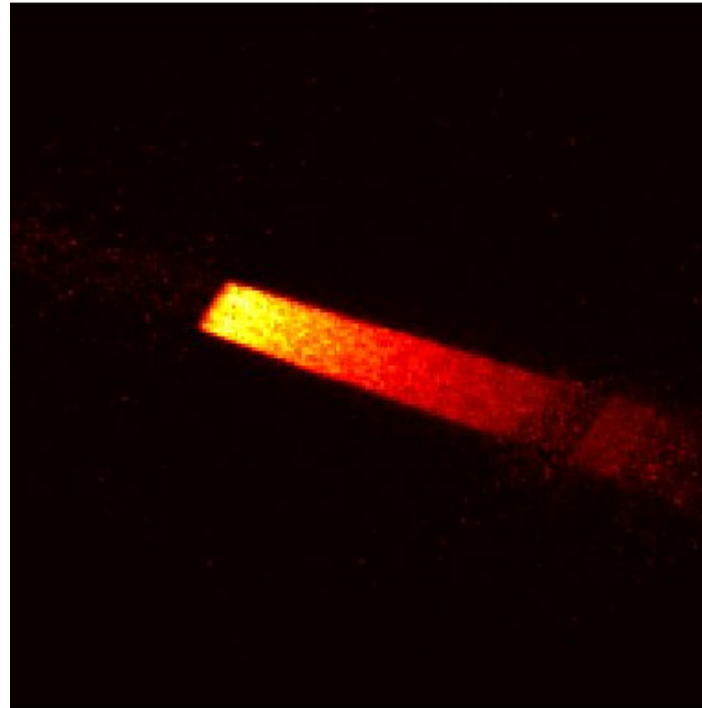
Neural Network Prediction

~9500 times faster than MC simulations

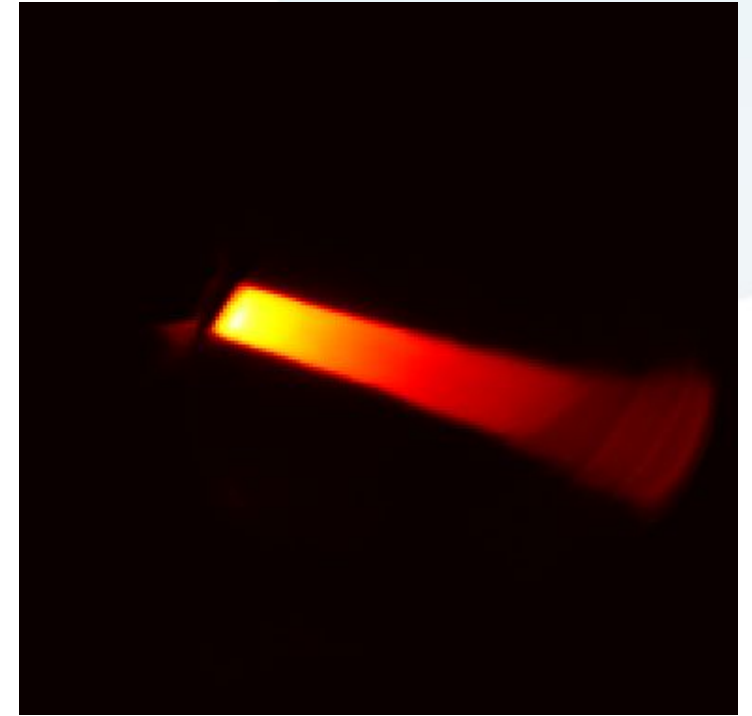
Results: Dose Prediction Fidelity



Patient CT



MC simulation



NN result

$$\Gamma 3\%/3\text{mm} = 60.8\%$$

Conclusions & Future Outlook

- ✓ **Proven Surrogacy:** TransUNet architecture successfully emulates complex photon transport physics in heterogeneous CT phantoms.
- ⚡ **Real-Time Potential:** Inference times below 100ms enable the possibility of online adaptive radiotherapy re-planning.
- 🧪 **Next Milestone:** Integration of **Graph Neural Networks (GNN)** for differentiable inverse planning and MLC fluence optimization.
- 🔬 **Scaling:** Expanding the training dataset beyond prostate cases to multi-organ cancer cases.



Thank you for your attention

Asier López Mantecón | alm223@alumnos.unican.es

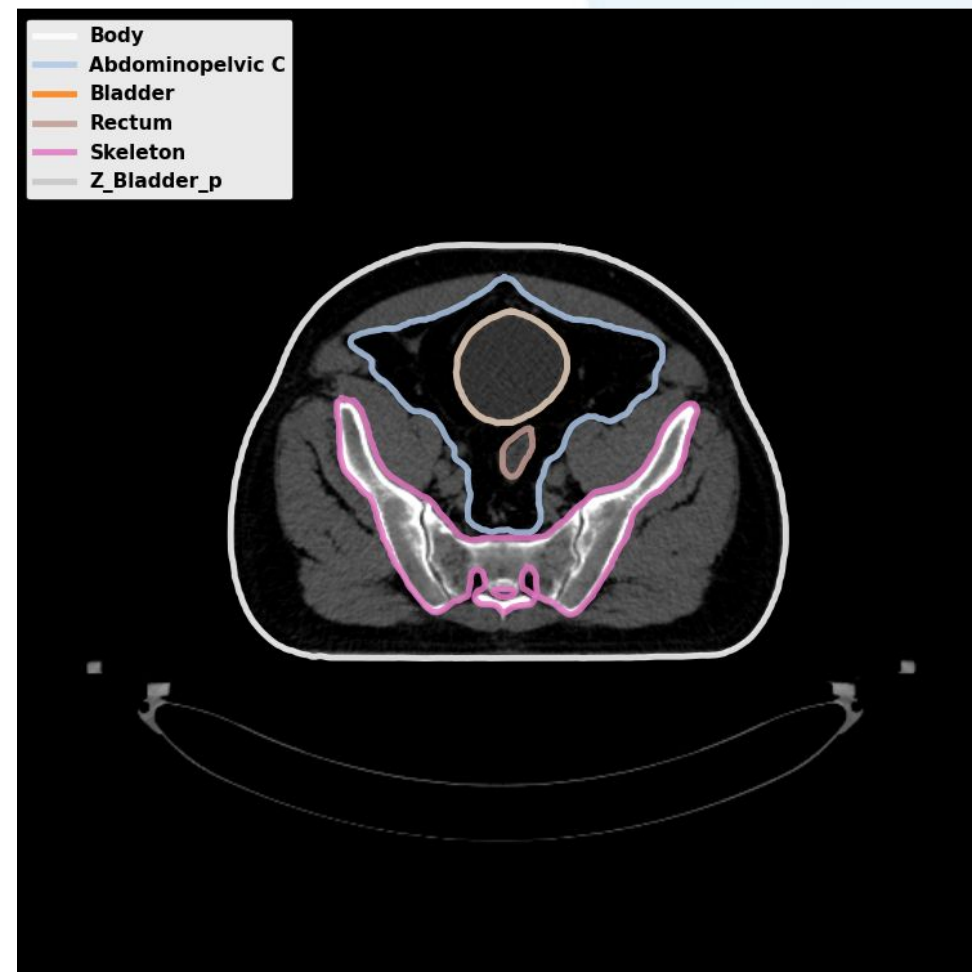
FlipPhysics 2026 – Valencia, Spain

GT Simulations: Physical Engine

OpenGATE v10.0.3 Framework

We implemented a robust radiation transport chain using the GEANT4-based OpenGATE engine. Key parameters include:

-  **G4EmStandardPhysics_option4:** Optimized for medical physics accuracy.
-  **30% Speed Increase:** Balanced performance compared to BERT_EMZ.
-  **High Precision:** 1mm production cuts for e^- , e^+ , and γ .



Neural Network Training Metrics

Optimization Functions

Training utilized a composite loss function to ensure physical and medical realistic accuracy

 **Clinic:** Ensures typical medical standards.

 **Geometry:** Controls spatial dose distribution.

 **Fidelity:** Focuses on deposited dose in organs.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{clinic}} + \mathcal{L}_{\text{geometry}} + \mathcal{L}_{\text{fidelity}}$$